QUANTITATIVE LIPIDOMICS OF HEALTHY AND DEMYELINATED MOUSE BRAIN BY MULTINUCLEAR NMR AND MASS SPECTROMETRY

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ABSTRACT

Myelin is a highly specialized multilamellar membrane that wraps around axons in the central nervous system and is essential for rapid nerve conduction and neuronal survival. Owing to its exceptionally high lipid content, composed mainly of phospholipids, glycolipids and cholesterol, alterations in myelin composition are associated with numerous demyelinating and neurodegenerative disorders, including multiple sclerosis. Despite its biological importance, the quantitative lipid composition of myelin and its changes during demyelination and remyelination remain incompletely characterized.

To address this question, we developed a quantitative analytical workflow combining **multinuclear NMR spectroscopy (^1H, ^31P and ^2H)** with **LC-MS/MS lipidomics** to characterize lipid composition in whole brain, corpus callosum and myelin-enriched fractions from mouse brain. The complementary nature of these techniques enables both absolute and relative quantification of major lipid classes together with detailed molecular lipid profiling.

The approach was applied to healthy mice, the **cuprizone model**, which reproduces reversible demyelination followed by remyelination, and the **shiverer mouse**, a genetic model of congenital hypomyelination caused by the absence of myelin basic protein (MBP). Quantitative comparison of these models reveals distinct alterations in phospholipid and cholesterol composition associated with myelin loss and recovery. In parallel, solid-state ^2H NMR performed on reconstituted lipid membranes provides complementary information on membrane organization and lipid dynamics.

Overall, this integrated strategy provides a comprehensive framework for investigating myelin lipid composition and membrane organization under physiological and pathological conditions and establishes a quantitative platform for studying myelin-related diseases.

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